Pseudo code and table comparison of time complexity:

```
Algorithm 1: findThirdLargest(array)
   Input: array of integers
   Output: the third largest value

   currentMax ← array[0]
   indexFirst ← 0

   for i ← 1 to array.length - 1 do
      if array[i] > currentMax then
         currentMax ← array[i]
         indexFirst ← i

   currentMax ← 0
   indexSecond ← 0

   for i ← 0 to array.length - 1 do
      if i ≠ indexFirst and array[i] ≥ currentMax then
         currentMax ← array[i]
         indexSecond ← i

   currentMax ← 0

   for i ← 0 to array.length - 1 do
      if i ≠ indexFirst and i ≠ indexSecond and array[i] ≥ currentMax then
         currentMax ← array[i]

   return currentMax
```

```
Algorithm 2: findThirdLargest(array)
   Input: array of integers
   Output: the third largest value

   max ← 0
   preMax ← 0
   prePreMax ← 0

   for each num in array do
      if num > max then
         prePreMax ← preMax
         preMax ← max
         max ← num
      else if num > preMax then
```

```
        prePreMax ← preMax
        preMax ← num
    else if num > prePreMax then
        prePreMax ← num

return prePreMax
```

Algorithm 3: findThirdLargest(array)

```
Input: array of integers
Output: the third largest value

top3 ← new OrderedSet()

for each num in array do
    top3.add(num)
    if top3.size() > 3 then
        top3.removeSmallest()

return top3.smallest()
```

	Best case	**Average case**	**Worst case**
Algorithm 1 (three passes)	O(n)	O(n)	O(n)
Algorithm 2 (single pass, cascading)	O(n)	O(n)	O(n)
Algorithm 3 (OrderedSet)	O(n)	O(n)	O(n)

Consider the following functions to determine the relationships that exist among the complexity classes they belong to.

Functions	**Complexity Class**
1 | 10	$\Theta(1)$
log(log n)	$\Theta(\log \log n)$
ln n | log n	$n\Theta(\log n)$
n^(1/k) (k > 3)	$\Theta(n^{(1/k)})$
n^(1/3)	$\Theta(n^{(1/3)})$
n^(1/3) log n	$\Theta(n^{(1/3)} \log n)$
n^(1/2)	$\Theta(n^{(1/2)})$
n^(1/2) log n	$\Theta(n^{(1/2)} \log n)$
n	$\Theta(n)$
n log n | log n^n	$\Theta(n \log n)$
n^2	$\Theta(n^2)$
n^3	$\Theta(n^3)$
n^k (k > 3)	$\Theta(n^k)$
2^n	$\Theta(2^n)$
3^n	$\Theta(3^n)$
n!	$\Theta(n!)$
n^n	$\Theta(n^n)$